



HK IMO Training 2022-2023
Problem Set 1
For 24 September 2022



Problem 1. Let $ABCD$ be a cyclic quadrilateral with circumcentre O . Let E be a point on the line AB such that $EB = EC$. Let F be a point on the line AD such that $FD = FC$. Prove that the circumcentre of $\triangle AEF$ lies on the perpendicular bisector of CO .

Problem 2. Solve the equation $m^3 + n^3 = m^2 + 42mn + n^2$ in positive integers.

Problem 3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{Z}$ such that

$$f(f(y) - x)^2 + f(x)^2 + f(y)^2 = f(y) + 2f(y)f(f(y))$$

for any $x, y \in \mathbb{R}$.

Problem 4. Two players X and Y play a game on an infinite grid as follows. Starting from X , the two players take turns choosing an undirected edge of the grid and assign a direction to this edge (in one of the two directions parallel to this edge). Player Y wins if there exists a directed cycle in the grid. Determine if Y has a winning strategy.